2ND SEMESTER LAB REPORT

TITLE : ANALYSIS OF SERIES AND PARALLEL CIRCUITS



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Submitted To:

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Lab Engineer

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EXPERIMENT NO 4

Lab Title: VERIFCATION OF OHM'S LAW

Objective: 1. **ANALYSIS OF SERIES AND parallel combinations of resistors**

Student Name: Muhammad Waqas -Muhammad Zeesha -Faiq Ali

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 10/March/2025 Signature:

LABORATORY EXPERIMENT NO. 7

ANALYSIS OF SERIES AND PARALLEL C IRCUITS

OBJECTIVE:

LAB TASKS:

TASK 1:

1. Solve using Nodal Analysis:

a) Take four resistors: two of 4.7kΩ, one of 6.8kΩ and one of 1kΩ. Calculate the resistance of each resistor, using the Resistor Colour Codes. Record the values in Data Table.

b) Now construct the circuit as shown in the figure. After verifying that the circuit is properly patched on the breadboard, switch on the trainer.

c) Using nodal analysis solve, solve for the node voltages. Record the values and calculations. Measure the node voltages using DMM. Record the values in data table.

d) Now turn off the power of trainer and DMM.

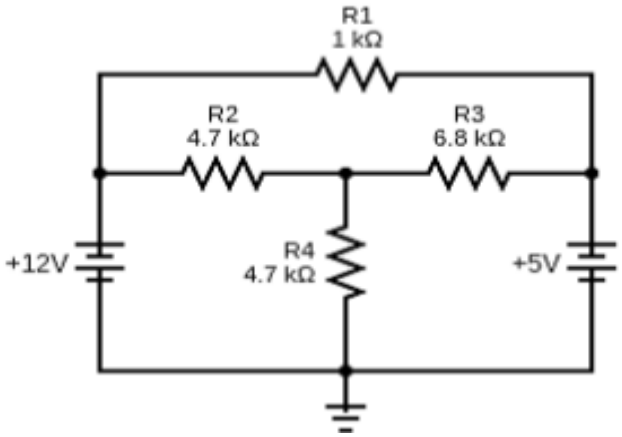


Table:

Nodes	V(Calculated)	V(Measured)
Node 1	12.0V	12V
Node 2	5.24V	5.74V
Node 3	5V	5V
Node 4	0V(Ground)	0V

Theoretical Calculations:

(12 - V) / 4.7 = (V - 5) / 6.6 + V / 4.7

V - 5) / 6.6 + V / 4.7

= ((V - 5) × 4.7) / (6.6 × 4.7) + (V × 6.6) / (4.7 × 6.6)

= (4.7V - 23.5) / 31.02 + (6.6V) / 31.02

= (4.7V - 23.5 + 6.6V) / 31.02

= (11.3V - 23.5) / 31.02

Final equation:

(12 - V) / 4.7 = (11.3V - 23.5) / 31.02

(12 - V) × 31.02 = (11.3V - 23.5) × 4.7

372.24 - 31.02V = 53.11V - 110.45

372.24 + 110.45 = 53.11V + 31.02V

482.69 = 84.13V

V = 482.69 / 84.13

Voltage:

V ≈ 5.74

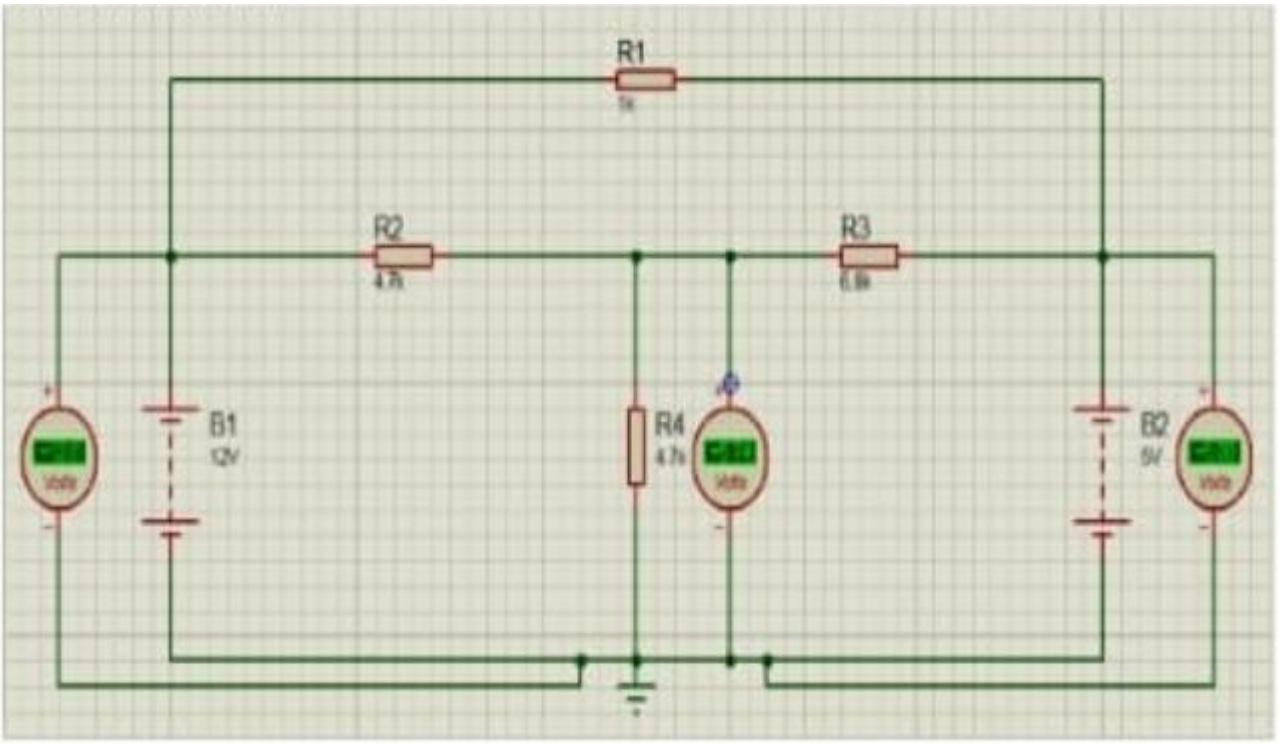
Verification:

Left side: (12 - 5.74) / 4.7 = 6.26 / 4.7 ≈ 1.33

Right side: (5.74 - 5) / 6.6 + 5.74 / 4.7 = 0.74 / 6.6 + 5.74 / 4.7 ≈ 0.11 + 1.22 ≈ 1.33

Therefore, V ≈ 5.74 is the solution to the equation.

Protius Simulation:



Conclusion:

NoDal analyis and KCL help us fgure out how electricity moves in circuits. KCL simply means that electricity coming into a point equals electricity going out Nodal analysis uses this idea to find voltages at different spots (nodes) in a circuit.

We write simple math equations showing how electricity flows between these spots. This lets us find unknown voltages anywhere in the circuit. These methods give us an easy step-by-step way to solve even tricky circuits with many parts.

Instead of trying to solve the whole circuit at once, we look at one spot at a time, using KCL at each point. This turns complex circuits into basic math problems that engineers can solve when building things like phones or electrical systems.